

Homework 4

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Introduction:

This report analyzes survey data from students in the CMSC 320: Introduction to Data Science class at the University of Maryland. The survey asked questions about the lifestyle and study habits of students for a particular exam in the class, and also included the total score the students received on the exam as well as the max points for the exam. The goal was to analyze whether the survey questions could be used to predict academic performance. After cleaning the data, I built three predictive models, which were a pass/fail classifier to predict whether a student passed or failed the exam, a letter-grade classifier to predict what exact grade the students got, and a regression model that predicts the numeric percentage grade. I used accuracy to evaluate the classification models and Mean Absolute Error (MAE) for the regression model. The results showed that there was a reasonable accuracy for the pass/fail classifier, a low accuracy for the letter-grade classifier, and a somewhat good MAE for the regression model.

Background:

The raw dataset for this report included survey responses submitted by students in the CMSC 320 class, as well as their scores for the exam. The survey questions asked about topics such as studying habits, sleep routines, lecture attendance, and thoughts about the exam. Every row in the dataset represented one student's set of responses and their score on the exam, as well as the total points on the exam. The dataset also included a timestamp column to see when each student submitted their response. These survey responses served as the raw data for the study. The goal was to transform the raw input data into a clean dataset that could be used to train machine learning models.

Data Cleaning:

Before any machine learning models could be trained on the dataset, the raw data had to be cleaned first. I did this by first converting the timestamp column to a proper datetime format, which would be easier to read if needed. After this, I dropped rows with a missing or zero value for the total score or max points column, as these fields are required to calculate the grade percentage. I also filtered out any rows where max points was less than or equal to 0, as this would cause an error when calculating the grade percentage. After this, I calculated the grade percentage for each student by dividing the total score by the max points and multiplying by 100. Using standard grading cutoffs, I converted these numeric percentages into letter grades, and then created a pass/fail label where A,B, and C grades were labeled as passes (1) and D and F grades were labeled as fails (0). To construct the features for the models, I dropped columns that would not be useful, including the timestamp, total score, max points, and the "extra credit"

question. Additionally, I dropped the target variables which were the letter grades, pass/fail variable, and grade percentages to prevent data leakage. The remaining survey response features had both categorical and numerical answers. The numerical answers were already in a usable form, but I had to convert the categorical columns into numerical indicator variables using one-hot encoding to get them in a usable form. After this, any missing values in the dataset were filled using the mean of each column, and this was done to ensure that the final feature matrix contained only clean data values that could be used to train the models.

Procedure:

After cleaning the dataset and using feature engineering to create the feature matrix for the machine learning models, I conducted three separate predictive tasks to evaluate how well the student survey responses could predict the academic performance of the students. All of the models were trained on 90% of the data and tested on the remaining 10% of the dataset. For the first task, I built three classification models, which were Logistic Regression, K-Nearest Neighbors with K being equal to 5, and a Decision Tree classifier. I used these three classification models to predict whether a student would pass or fail the exam, and the accuracy on the testing set was used as the evaluation metric for the models. The second task involved predicting the letter grades (A, B, C, D, F) of the students. I built three classification models for this, which were Logistic Regression, Random Forest classifier, and K-Nearest Neighbors with K being equal to 5. The performance of each model was again based on the accuracy on the testing set. Finally, for the third task I used a Linear Regression model to predict the exact percentage grade of students. I evaluated the performance of the model by using MAE which measures how far the predictions were from the correct grades on average.

Findings:

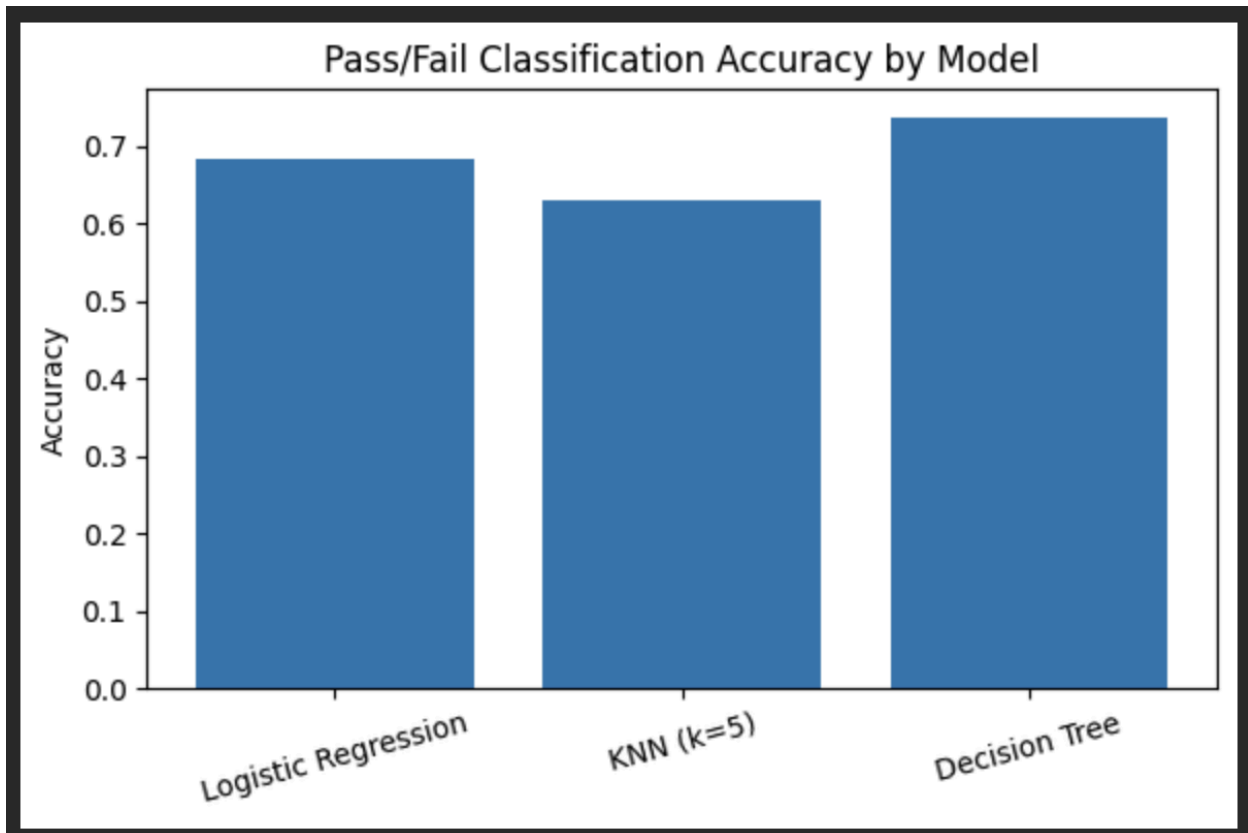
The three modeling tasks revealed several key insights about how the survey response can predict the academic performance of students. The results from the first task are displayed below, along with a bar graph showing a visual representation of the accuracies of each model.

Part One:

Logistic Regression accuracy: 0.684

KNN (k=5) accuracy: 0.632

Decision Tree accuracy: 0.737



When predicting the pass/fail outcomes, the Decision Tree model performed the best with an accuracy of 0.737, meaning it correctly predicted the output nearly 74% of the time. This slightly outperformed both the Logistic Regression model (0.684), and the K-Nearest Neighbors model (0.632). The bar graph clearly shows that the decision tree model has the highest accuracy performance. These results show that the pass or fail outcomes are reasonably predictable from the survey features, as all three models achieved an accuracy of at least 63%.

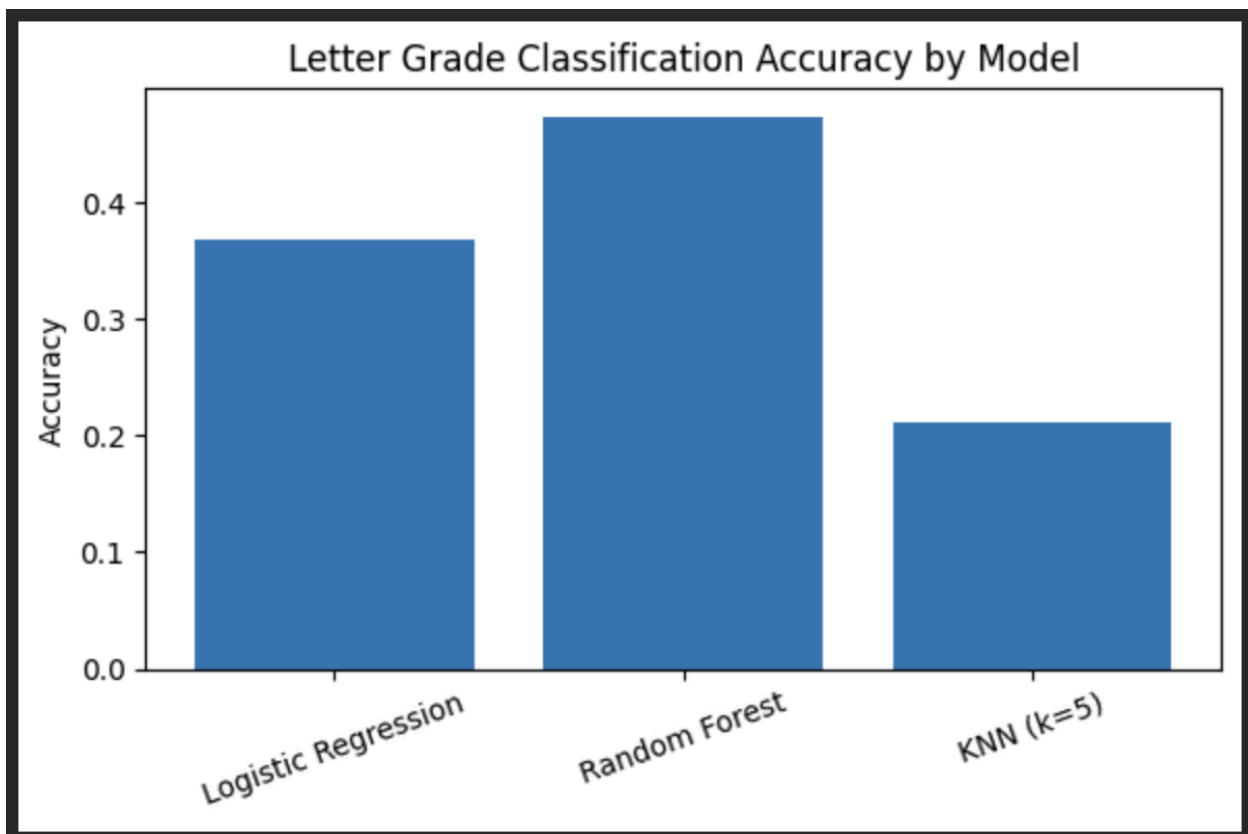
The results from the second task are displayed below, where each model attempted to classify students into exact letter grades. A bar graph showing the model accuracies for each model is also displayed, so that the accuracies can be compared visually.

Part Two:

Logistic Regression accuracy: 0.368

Random Forest accuracy: 0.474

KNN (k=5) accuracy: 0.211

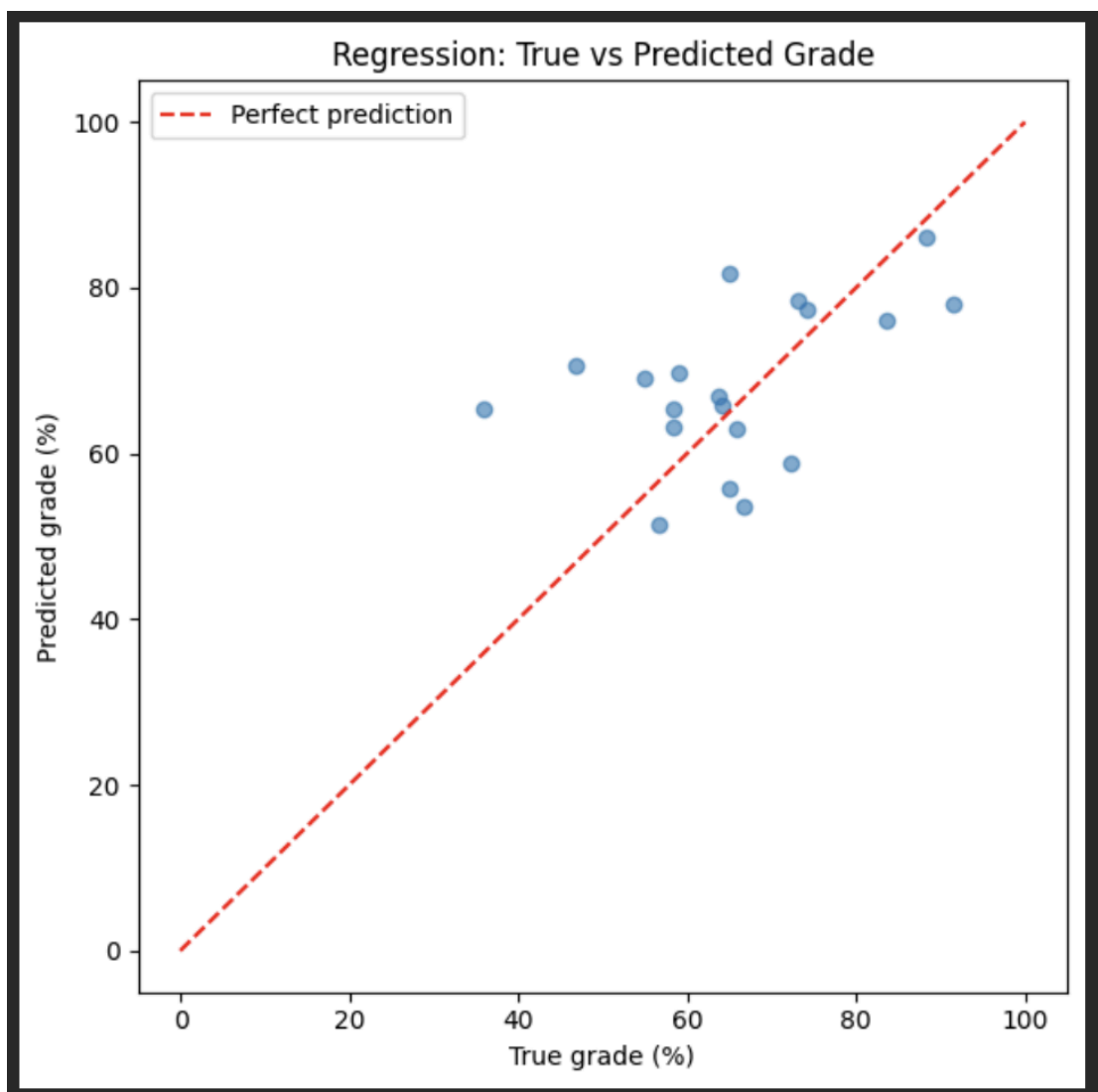


From these results, it is evident that the prediction performance declined significantly when attempting to classify the students into exact letter grades. The bar graph clearly shows how the best performing model is the Random Forest classifier. When looking at the results, the Random Forest model had an accuracy of 0.474, which was much higher than the accuracy of the Logistic Regression model (0.368) and the accuracy of the K-Nearest Neighbors model (0.211). Although the Random Forest model performed the best, its accuracy of 0.474 is still not very high, and all

of the models had a low accuracy with the K-Nearest Neighbors model performing especially poorly. These low values indicate that distinguishing between actual letter grades is much more challenging than distinguishing between pass or fail. This is likely because of the fact that differences between letter grades are small and may not be captured by the survey responses.

The results from the regression task are displayed below, where a Linear Regression model was used to predict the actual percentage grade students scored on the exam. The scatter plot showing the regression results is also displayed below.

Part 3:
Linear Regression MAE: 9.86 percentage points



From these results, it is shown that the Linear Regression model produces a Mean Absolute Error of 9.86 percentage points, which means that on average the predictions were off by about ten points from the true grade. This is also shown in the scatter plot of the actual grades compared to the predicted grades. In the plot, each point represents a student's true score on the x-axis, and the model's predicted score on the y-axis. The red dashed line represents perfect prediction, where the predicted grade would match the actual grade exactly. Many points are shown to fall above or below this line, indicating decently large prediction errors for some students. However, some predictions also lie close to the line, suggesting reasonable prediction accuracy for these students. This shows that the regression model struggles to produce exactly precise grade predictions, which reinforces the idea that exact numeric percentages are difficult to predict from just the survey responses. However, the model performs decently well since the MAE is 9.86 percentage points, which is very reasonable.

Conclusion:

In conclusion, the report demonstrates that using the student survey responses dataset to predict the academic performance of students has varying results, depending on what is being predicted. When trying to classify the students based on if they passed or failed the exam, the machine learning models achieved the highest performance. This shows that the survey responses can be used to somewhat accurately predict if a student passed or failed the exam. On the contrary, the models struggled with the letter grade classification task, where all three models achieved low accuracy. This indicated that models trained on the survey responses alone cannot distinguish between different letter grades. Finally, the regression model achieved a decent Mean Absolute Error, further showing that it is difficult to predict exact grades. In summary, although the survey responses can help reveal general academic performance patterns, it is difficult to accurately predict precise grades with the survey data. Therefore, to achieve more reliable predictions, machine learning models must rely on more comprehensive datasets to effectively predict the academic performance of students.